

Quick Review

In most circumstances, a basis with vectors orthogonal to one another, or an *orthogonal basis*, is hard to come by. Even rarer is an *orthonormal basis*, or a basis in which every vector has a norm of 1 while also being orthogonal to every other vector in the basis.

However, last week you learned that the only condition that needs to be met for orthogonality between two vectors u and v is that

$$\langle u, v \rangle = 0,$$

where $\langle u, v \rangle$ is any inner product.

Once all vectors in a basis are orthogonal, to obtain an orthonormal basis we only need to *normalize* all the vectors v_i . In this section we'll denote the normalized vector as q_i such that

$$q_i = \frac{v_i}{\|v_i\|} = \frac{v_i}{\sqrt{\langle v_i, v_i \rangle}}.$$

However, when we do not have an orthogonal basis we must resort to other methods to obtain an orthonormal basis. The most efficient way (though it still takes forever!) to produce an orthonormal basis from *any* set of vectors, is to use the *Gram-Schmidt process*. Below is an outline of the process.

Let $S = \{u_1, u_2, \dots, u_r\}$ be any basis for W .

Gram-Schmidt Process

1. Select any vector u_1 to be your v_1 .
2. Normalize v_1 to get q_1 .
3. Find v_2 by performing $v_2 = u_2 - \text{proj}_{q_1} u_2$.
4. Normalize v_2 to get q_2 .
5. Find v_3 by finding the component of u_3 that is orthogonal to $\text{span}(\{q_1, q_2\})$. This may be written as

$$v_3 = u_3 - \text{proj}_{\text{span}(\{q_1, q_2\})} u_3 \quad \text{OR} \quad v_3 = u_3 - \text{proj}_{q_1, q_2} u_3$$

6. Normalize v_3 to get q_3 .
7. Repeat this process until you have found q_r , or in other words, you have r orthonormal vectors.
8. Your orthonormal basis will then be $\{q_1, q_2, \dots, q_r\}$.

As an extension of the Gram-Schmidt process, QR factorization is a way to relate the original basis $\mathbf{A} = [u_1 \mid u_2 \mid \dots \mid u_r]$ to the new, orthonormal one (using $\mathbf{A} = \mathbf{QR}$).

In short, $\mathbf{Q}$ will be the matrix defined by $[q_1 \mid q_2 \mid \dots \mid q_r]$. $\mathbf{R}$ will be the matrix such that

$$R = \begin{bmatrix} \langle u_1, q_1 \rangle & \langle u_2, q_1 \rangle & \cdots & \langle u_n, q_1 \rangle \\ \langle u_1, q_2 \rangle & \langle u_2, q_2 \rangle & \cdots & \langle u_n, q_2 \rangle \\ \vdots & \vdots & & \vdots \\ \langle u_1, q_n \rangle & \langle u_2, q_n \rangle & \cdots & \langle u_n, q_n \rangle \end{bmatrix}$$

This can be further simplified by thinking about the Gram-Schmidt process. For example, are the vectors u_1 and q_2 orthogonal? They are! It turns out everything below the diagonal is orthogonal, so all the terms will become 0!

Problems

1. Let the inner product on $\mathbb{P}_2$ have the definition shown below. (**Adapted from Ch 6.3 Problem #43**)

$$\langle p, q \rangle = \int_0^1 p(x)q(x)dx$$

- (a) Let $p(x) = x^2 + 3$ and $q(x) = x - 3$. Evaluate $\langle p, q \rangle$. Are the two vectors orthogonal?

- (b) What are the two conditions that *must* be satisfied for a basis to be an orthonormal basis?

(1) _____

(2) _____

- (c) Using the Gram-Schmidt process transform the standard basis of $\mathbb{P}_2$ into an orthonormal basis.

(d) Find the QR-decomposition of the orthonormal basis from part (c).

2. Let there be a set $S = \{(1, 2, 3), (0, 2, 0), (2, 2, 6)\} \in \mathbb{R}_3$ where the inner product is that of the standard inner product of $\mathbb{R}_3$.

(a) Find an orthonormal basis for the set S .

(b) Based on your result from part (a), what can you say about the linear independence of S ?

3. **(PROOF).** Prove that if W is a subspace of a finite-dimensional vector space V , then the mapping $T : V \rightarrow W$ defined by $T(v) = \text{proj}_W v$ is a linear transformation. **(Ch 6.3 Problem #55a)**

4. Let $\mathbf{A}$ be the matrix defined below, and x and y be vectors in $\mathbb{R}_2$.

$$\mathbf{A} = \begin{bmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \quad x = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \quad y = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$$

- (a) Find $\det(\mathbf{A}^T \mathbf{A}^{-1})$.

- (b) Find $\mathbf{A}x \cdot \mathbf{A}y$.